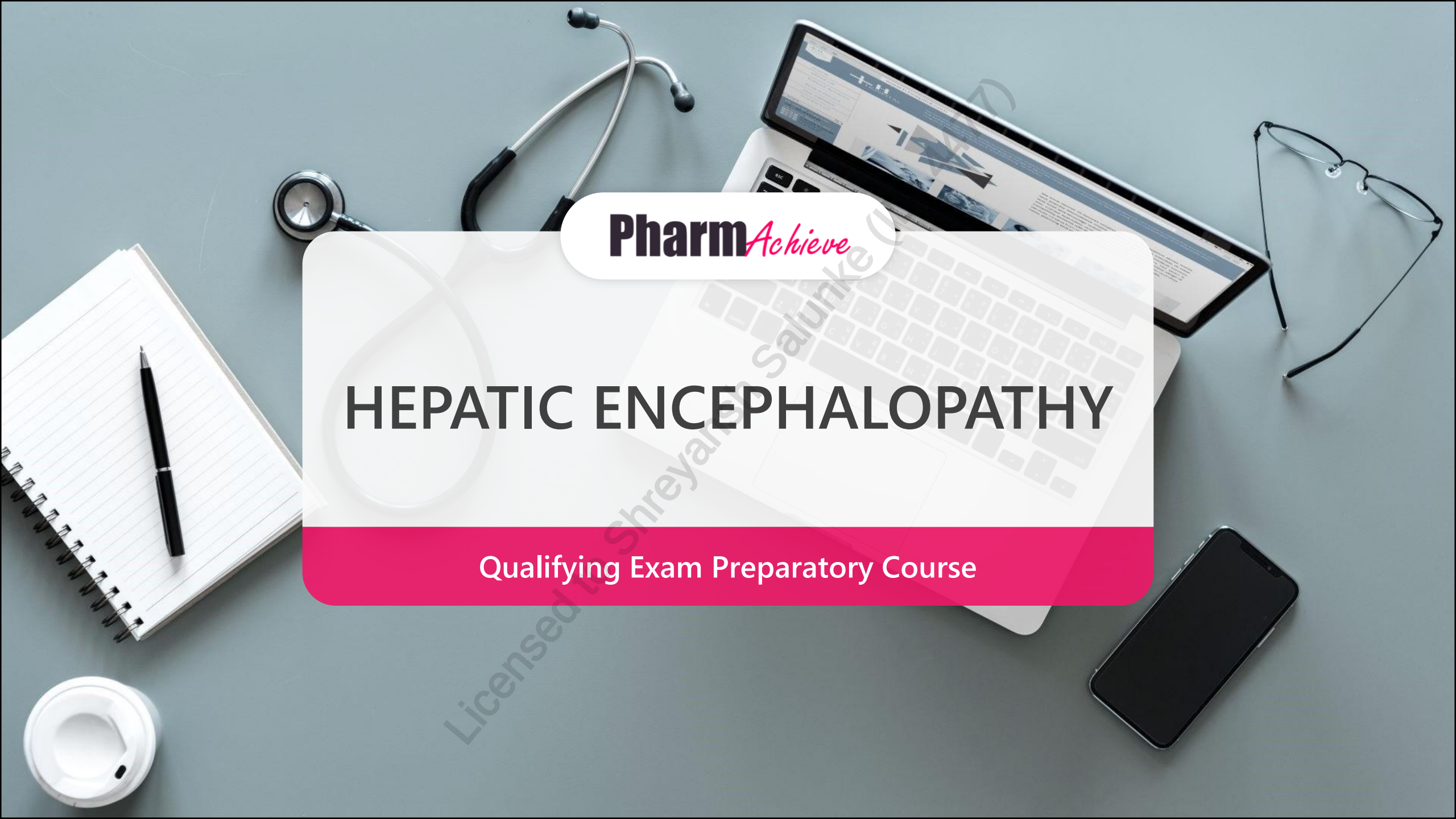




Pharm*Achieve*

HEPATIC ENCEPHALOPATHY

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

AASLD American Association for the Study of Liver Diseases

ADR Adverse Drug Reaction

BBB Blood Brain Barrier

BID Bis In Die (Twice Daily)

BM Bowel Movement

CNS Central Nervous System

CT Computed Tomography

DNA Deoxyribonucleic Acid

ED Emergency Department

GABA Gamma-Aminobutyric Acid

GCS Glasgow Coma Scale

GI Gastrointestinal

H Hours

HE Hepatic Encephalopathy

ICU Intensive Care Unit

INR International Normalized Ratio

IV Intravenous

K⁺ Potassium

LFT Liver Function Tests

MAP Mean Arterial Pressure

MAP Mean Arterial Pressure

MOA Mode of Action

MRI Magnetic Resonance Imaging

N/V/D Nausea/Vomiting/Diarrhea

Na⁺ Sodium

NH₃ Ammonia

NH₄⁺ Ammonium

NMDA N-methyl-D-aspartate

NS Normal Saline

PO Per Os (Oral)

PR Per Rectum (Through the Rectum)

PRN Pro Re Nata (As Needed)

Q Quaque (Every)

QAM Quaque Die Ante Meridiem (Every Morning)

QHS Quaque Hora Somni (Every Night At Bedtime)

QID Quater In Die (Four Times A Day)

RNA Ribonucleic Acid

SNRI Serotonin-Norepinephrine Reuptake Inhibitor

TID Ter In Die (Three Times A Day)

PATHOPHYSIOLOGY

- **Reversible impairment of brain function** due to toxin and fluid buildup
 - Impairment of brain function ranges from minor neurologic deficits to coma and death
- Exact mechanism is **unknown**: liver failure leads to accumulation of toxins that enter the brain
- **Ammonia** is implicated
 - Ammonia (NH_3) comes from the GI tract when bacteria breakdown proteins
 - The liver normally prevents ammonia from entering systemic circulation by converting it to glutamine
- Liver dysfunction results in **decreased clearance of ammonia**
 - Hyperammonemia enhances **uptake of amino acids into brain**, affecting neurotransmitter synthesis
 - Astrocytes metabolize ammonia into glutamate, **increased intracellular osmolarity**, which may lead to **vasodilation and brain edema**
- **Other proposed mechanisms**: enhanced GABA transmission, benzodiazepine/NMDA receptors, altered levels of catecholamines, alteration of blood brain barrier, hyponatremia, inflammation caused by infection

CLINICAL PRESENTATION

Insomnia or hypersomnia

Typically precedes neurologic signs

Psychiatric features

Euphoria/depression, disorientation, inappropriate behaviour, confusion/delirium

Focal neurological deficit possible

Hemiparesis (one-sided muscle weakness)

Neurological features

Bradykinesia, slurred speech, confusion, altered focus, nystagmus

Advanced encephalopathy

Loss of consciousness, coma

RISK FACTORS

- HE may be precipitated by:
 - Acute liver failure (type A HE)
 - Hepatic bypass/shunt (type B HE)
 - Hepatic cirrhosis (type C HE)
 - Increased protein load (GI bleed, excess dietary protein, constipation)
 - Infection
 - Constipation
 - Hypokalemia: increases renal ammonia production
 - Alkalosis: promotes NH_4^+ (unable to cross BBB) conversion to NH_3 (readily crosses BBB)
 - Dehydration (vomiting, diarrhea, hemorrhage, diuretics)
 - Portal/hepatic vein thrombosis

DIAGNOSIS

FYI

- **Psychometric tests** to check mental status changes (psychometric hepatic encephalopathy score)
- **Physical findings of severe liver disease:** muscle wasting, jaundice, ascites, palmar erythema, edema, spider telangiectasis
- **Laboratory testing** for blood ammonia concentration, hepatic abnormalities (e.g. LFTs), electrolyte disturbances (Na^+ , K^+), and intoxication
- **CT/MRI** to rule out other causes (e.g. hematoma) and to observe brain edema
- **Glasgow Coma Scale (GCS)** in patients with significantly altered consciousness (assesses eyes, verbal, and motor responses)
 - A higher score (max. 15) indicates better neurological function and alertness, while a lower score (min. 3) indicates more severe impairment of consciousness or coma
- **Child-Pugh** or **Child-Turcotte Score** is used to assess prognosis of chronic liver disease (cirrhosis)
 - Includes 1 to 3 points for each of the following: total bilirubin, serum albumin, INR, ascites, hepatic encephalopathy
 - **Class A:** score of 5-6 with a 100% 1-year survival and 85% 2-year survival
 - **Class B:** score of 7-9 with a 80% 1-year survival and 60% 2-year survival
 - **Class C:** score of 10-15 with a 45% 1-year survival and 35% 2-year survival

MEDICATION-RELATED CAUSES

Benzodiazepines

- Increases GABA signaling worsening or precipitating symptoms of HE

CNS depressants

- Opioids/narcotics
- Alcohol

GOALS OF THERAPY

- ✓ Resolve acute episode of hepatic encephalopathy
- ✓ Correct precipitating factors:
 - ✓ Hypovolemia (due to vomiting, diarrhea, diuretics, 3rd spacing with large volume paracentesis)
 - ✓ Infection
 - ✓ GI hemorrhage
 - ✓ Hypokalemia
 - ✓ Constipation
 - ✓ Portal or hepatic vein thrombosis
 - ✓ Medications worsening symptoms (narcotics, benzodiazepines, alcohol)
- ✓ Prevent future episodes/complications
- ✓ Manage portal hypertension
- ✓ Decrease serum ammonia levels

NON-PHARMACOLOGICAL MEASURES

1. Treat/eliminate other underlying risk factors

- **Infection, alkalosis, GI bleed, constipation**
 - All possible factors for increased ammonia levels
- **Dehydration, electrolyte imbalance, hypoxia**
 - May worsen edema and cognitive function
- **Precipitating medications**
 - Benzodiazepines, narcotics, alcohol

2. Restrict protein intake

- **Short term measure** only if necessary (as protein restrictions may be associated with increased mortality)
 - Chronic protein restriction can lead to muscle wasting
 - Many patients with liver dysfunction are malnourished and have higher than normal protein needs
- HE patients should have a **protein intake of 1.2-1.5 g/kg ideal body weight/day**

PHARMACOLOGIC MEASURES

- **Treatment strategy**

- **Lactulose** – titrate dose to keep 2-3 loose bowel movements/day
- If no improvement in 24-48 hours, add **rifaximin** or **metronidazole***
- **Flumazenil** (benzodiazepine antagonist) is **not recommended** for routine clinical use

- **Prophylaxis strategy**

- **Lactulose** – started/continued after first episode HE
- **Rifaximin** – started/continued after second episode of HE

*Metronidazole Level II-3 evidence, Grade B Recommendation- 2014 AASLD Guideline

PHARMACOLOGIC MEASURES

LACTULOSE

MOA	- Metabolized by colonic bacteria into lactic acid and acetic acid - Converts ammonia (NH_3) into ammonium (NH_4^+) which is ionized and not absorbed resulting in lowering of blood ammonia concentration
Efficacy	More effective than placebo for improving symptoms, but no benefit on mortality
Dose	25-45 mL PO Q1-2H, then titrate to BID-QID (target to 2-3 loose BMs/day)
Interactions	None
Side Effects	Bloating, flatulence, cramps, diarrhea
Comments	Can be given rectally (300 mL of lactulose in 700 mL water or NS, retained 30-60 min and given Q4-6H) if patient unable to take oral

PHARMACOLOGIC MEASURES

RIFAXIMIN

MOA	• Inhibits bacterial RNA synthesis by binding to RNA polymerase • Decreases activity of urease-producing bacteria which decreases ammonia production
Efficacy	• Little to no absorption in the GI tract
Dose	• 550 mg PO BID (empty stomach)
Interactions	• Oral contraceptives, sodium picosulphate, decreases warfarin anticoagulative effect
Side Effects	• Peripheral edema, dizziness, fatigue, ascites, nausea, nasopharyngitis, muscle spasms, prolonged use increases risk of <i>C. difficile</i>

PHARMACOLOGIC MEASURES

METRONIDAZOLE

MOA	- Disrupts DNA helical structure - Decreases activity of urease-producing bacteria which decreases ammonia production
Interactions	- May ↑ INR with warfarin, ↑ cardiotoxicity with amiodarone, ↑ toxicity of lithium, carbamazepine, cyclosporine, phenytoin, tacrolimus - Alcohol intake may result in disulfiram-like reactions
Side Effects	- Metallic taste, N/V/D, rash - Rare peripheral neuropathy (at high cumulative doses > 30 to 40 g), ototoxicity, leukopenia

MONITORING PARAMETERS

Efficacy Endpoints

Parameter	Degree of Change	Timeframe
Ammonia plasma level	23.5 to 41.1 $\mu\text{mol/L}$ Titrate to keep 2-3 BM/day	2-3 days
Confusion/loss of consciousness/ neurological or psychiatric symptoms	Improve Normalize	2-3 days 1-2 weeks

Safety Endpoints

Parameter	Degree of Change	Timeframe
ADR from lactulose/ metronidazole therapy	Prevent or minimize	Duration of therapy

THERAPEUTIC TIPS/TAKEAWAY

- ✓ Hepatic encephalopathy presentation consists of neuromuscular and cognitive dysfunction
- ✓ Correct/treat precipitating risk factors (e.g. alcohol, GI bleed, infection, hypokalemia)
- ✓ Ensure lactulose is not discontinued in patients presenting with hepatic encephalopathy once improved. **Aim to keep 2-3 BM per day and titrate dose of lactulose** accordingly
- ✓ Initiate rifaximin in addition to lactulose **after 24-48 hours, if no improvement in mental status** is seen during an acute episode

CASE

DF is a 63-year-old male presenting to the emergency department with three days of increasing confusion and lethargy. On the day of presentation, he was found by his family in the morning unable to get out of bed, and with significantly impaired consciousness. He is now in the resuscitation area and is being evaluated for hepatic encephalopathy (HE). Vital signs are within normal limits. DF has a past medical history significant for liver disease with cirrhosis secondary to ethanol and hepatitis C (Child-Pugh C) and gynecomastia as well as depression. He has not had any prior episodes of HE. He currently takes venlafaxine 225 mg PO daily, furosemide 40 mg PO QAM, spironolactone 200 mg PO daily, mirtazapine 15 mg PO QHS PRN, nadolol 40 mg PO daily, and vitamin D 1000 units PO daily. He has no known allergies. He lives with his family (wife, GF, and two adult children), he works as an artist (metalwork—semi-retired), and he averages four 12-ounce cans of beer/day, more on weekends.

1. DF has a current GCS of 11, is protecting his airway and has a CT scan showing a normal brain without evidence of hemorrhage, lesion or mass effect. What is the optimal approach to management for DF at this time?
 - a) Initiate metolazone 2.5 mg PO daily
 - b) Reduce nadolol to 20 mg PO daily
 - c) Initiate rifaximin 550 mg PO daily
 - d) Initiate lactulose 30 mL PO Q2H



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2. Over the next several hours, DF continues to have a decline in his level of consciousness with a GCS nadir of 8. He is intubated for airway protection and transferred to the ICU. In the ICU, his ammonia level returns as 55 mmol/L. He has received lactulose 30 mL total over the past 3 hours since ED presentation without a bowel movement. Which of the following is the best option to manage DF at this time?
- a) Initiate nitroglycerin infusion titrated to a MAP of 60
 - b) Increase lactulose to 30 mL PO Q1H PRN
 - c) Initiate metronidazole 500 mg IV Q8H
 - d) Initiate lactulose 300 mL in 700 mL NS PR BID



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3. On day 6 of his hospital stay, DF remains intubated in the ICU, however, his GCS has now improved to 14 and the team is preparing for extubation. His relevant HE medication includes lactulose 30 mL PO TID, rifaximin 550 mg PO daily and metronidazole 500 mg PO BID. His vital signs are stable and within normal limits. The ICU team asks what the best approach would be to reduce his chance for recurrent hepatic encephalopathy requiring hospital admission. What is the best response?
- a) Continue lactulose, rifaximin and metronidazole and restart nadolol
 - b) Stop rifaximin, continue lactulose and metronidazole and restart nadolol
 - c) Continue lactulose, stop metronidazole and rifaximin
 - d) Stop spironolactone, continue rifaximin and lactulose and restart furosemide



CASE

4. On day 12 of his hospital stay, DF is stable, with a GCS of 15 and no complaints. He is being discharged home on the following medications:

- Nadolol 20 mg PO daily
- Spironolactone 100 mg PO daily
- Furosemide 20 mg PO QAM
- Lactulose 30 mL PO TID
- Venlafaxine 150 mg PO QAM

You are counseling DF and his wife about his new medication regimen. Which of the following statements best reflect an accurate counselling point?

- a) The significant interaction between his nadolol and venlafaxine requires separation of doses by at least two hours
- b) Gynecomastia, likely due to his spironolactone will subside due to the reduced dose
- c) His lactulose dosing may change over the next few weeks as we are aiming for 2-3 bowel movements per day
- d) The reduction in venlafaxine dosing will lead to SNRI withdrawal headaches and confusion over the next several weeks



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CHANGE LOG

August 2025

- Formatting and grammar changes throughout
- Slide 3: Updated legend
- Slide 5: Added definition of hemiparesis
- Slide 7: Added Glasgow Coma Scale
- Slide 10: Added 'ideal body weight' and note on mortality
- Slide 15: Updated ammonia plasma level according to SI units
- Slide 21: Updated references

November 2025

- Formatting changes made throughout

March 2026

- Slide 2: Updated NAPRA competencies
- Formatting edits made, no content changes